

Title slide | Opening

Hello everyone, and thank you for being here.

Today, I will present my work on controlling a seven-joint robot for grinding. The goal is to help the tool align with a surface while reducing unwanted motion of the arm.

Slide 1 | Motivation

First, let me explain why tool alignment matters.

For grinding, the tool face needs to align with the surface. However, the real surface angle may differ from the angle used by the controller. Moreover, joint friction can also affect the tool orientation. The controller therefore allows the tool to rotate passively during contact, helping it align with the real surface. This video shows the idea.

[Play the full video: about 51 seconds. Then say:]

As you can see, the tool rotates even though its desired orientation remains fixed.

Slide 2 | Overview

With that goal in mind, here is the structure of the presentation.

I will first explain the controller, then show the contact experiments and their results. After that, I will discuss the extra joint motion and the null-space controller, before summarizing the main findings.

Slide 3 | Cartesian pose

To explain the controller, I will start with the tool pose.

The pose describes where the tool is and how it is rotated. It has three position coordinates and three orientation coordinates. Using forward kinematics, we calculate this pose from the seven measured joint angles.

Slide 4 | Surface frame

We can now describe the directions in which the tool interacts with the surface.

The normal points out of the surface, while the two tangents lie along it. These directions define the force and moment components, allowing us to choose where the tool should be compliant and where it should remain stiff.

Slide 5 | Cartesian impedance law

Cartesian impedance control gives this interaction the behaviour of a spring and a damper.

Reading the force equation, f equals K_p times e_p , plus D_p times the velocity error. Here, e_p is the desired position minus the measured position. The stiffness term therefore pulls the tool towards its target. The damping term uses desired minus measured velocity, so for a fixed target it opposes motion. If I push the tool and hold it still, the damping term vanishes, but the spring term remains because the position error remains. Once I release the tool, that spring term drives it back, while damping slows the return.

The moment equation follows the same pattern: m equals K_R times e_R , plus D_R times the angular-velocity error. Here, e_R is the rotation angle times its axis. Force and moment form the wrench F below. The zero off-diagonal blocks keep translation and rotation separate at the compliance reference. Lower stiffness allows more displacement under a given load, while damping reduces oscillation. During contact, lower rotational stiffness therefore lets the surface moment turn the tool more easily, even with a fixed desired orientation.

Slide 6 | Real-time control

To apply this wrench, the controller must convert it into joint torques.

The first equation, $\tau = J^T F$, maps the force and moment to all seven joints. In the feedback direction, $\dot{x} = J \dot{q}$ gives the tool velocity from joint velocities. We then add the null-space torque and Coriolis compensation to the Cartesian torque. However, gravity compensation is handled internally by the robot. This loop repeats every millisecond. After each update, we compare the reference and measured state again. If contact changes the tool pose, these errors change the wrench in the next cycle, so the response adapts continuously.

Slide 7 | Centre of compliance: a virtual reference

We can also choose where the virtual spring and damper act.

The tool centre point, or TCP, is our controlled point on the end effector, or EE. The centre of compliance, or CoC, is the virtual reference for stiffness and damping. Shifting this reference introduces a lever effect. The equation $m \ddot{a}$ at the TCP equals $r \times c \times f$ describes the added moment: it depends on both the displacement and force directions. For a force parallel to $r \times c$, the cross product is zero, so this added moment vanishes.

Slide 8 | Supporting and opposing CoC shifts

Whether that moment helps depends on the direction of the shift.

On the left, the added moment supports alignment, while on the right, it opposes alignment. The CoC shift must therefore be chosen to match the initial angular offset of the tool.

Slide 9 | Translation-rotation coupling

The equations here show how we introduce that coupling into the controller.

The displacement r_c runs from the TCP to the CoC. It defines the point-shift adjoint matrix A , written as Ad of r_c . We use it to transform both matrices: K at the TCP equals $A^T K_c A$, and D at the TCP equals $A^T D_c A$. The right-hand A maps errors and velocities to the CoC, and A^T maps the resulting wrench back to the TCP.

These transformed matrices generally have nonzero off-diagonal blocks. A position error can therefore create a moment, and an orientation error can create a force. The damping blocks similarly couple linear and angular velocity errors. This is how the CoC shift changes the force and turning response together.

We transform stiffness and damping separately. The coupling is between translation and rotation within each matrix. When the CoC coincides with the TCP, r_c is zero and A becomes the identity, recovering the original uncoupled matrices.

Slide 10 | Contact experiment

With the controller defined, I can now show the contact experiment.

I deliberately start with an angular offset to represent a mismatch between the assumed and real surface orientations. The robot then approaches, makes contact and performs the grinding motion. Motion along the normal and rotation about the two tangent axes are compliant, while the other directions remain stiff.

Slide 11 | Plausibility experiment

Before assessing alignment, I first checked the basic impedance response.

For this test, I pushed and rotated the tool by hand, as shown in the video.

[Play the full video: about 14 seconds. Then say:]

The next two plots compare the commanded and estimated values with the spring-law predictions. It is important to note that the estimated values are model-based measurements.

Slide 12 | Force plausibility

Let us first check the force against the spring term K_p times e_p .

A displacement of about twenty millimetres predicts approximately minus twenty newtons. The commanded mean is almost the same, while the model estimate is slightly larger. Both means below the plot use the shaded steady-state interval.

Slide 13 | Moment plausibility

The rotational test checks the corresponding term K_R times e_R .

A rotation of about six and a half degrees gives predicted and commanded mean moments of roughly one point seven newton metres. The estimated mean is slightly higher. Again, the means use the shaded interval. Together, these tests support the expected spring behaviour.

Slide 14 | Moment during contact alignment

We can now look at the wrench during actual contact alignment.

In this separate test, the force curves remain close. However, the model-estimated moment is larger during alignment because it includes the contact moment from the surface, which helps turn the tool. Once the motion settles, the values become close again.

Slide 15 | Effect of rotational stiffness

To understand this turning response, I first varied the rotational stiffness.

The plot shows the remaining angular error relative to the calibrated surface. Closer to zero means better alignment in this direction. Higher rotational stiffness leaves a larger error because it resists the rotation caused by contact.

Slide 16 | CoC shift and entry offset

Next, I varied the centre of compliance to change the coupling.

For a positive starting offset, a positive CoC shift helps alignment, whereas a negative starting offset benefits from a different CoC position. A larger shift is therefore not always better: its direction and size must match the initial angular offset.

Slide 17 | Turning response over time

The time histories show how this difference develops.

A supporting CoC shift helps alignment happen faster, while an opposing shift leaves much of the initial angular error. The normal forces are broadly similar, but the moments differ. In these tests, the CoC shift therefore mainly changes the rotational response.

Slide 18 | Null-space controller

So far, we have controlled the tool, while the null-space controller manages the extra motion of the arm.

With seven joints and six independent tool-pose coordinates, seven minus six leaves one extra degree of freedom. The condition $J \dot{q} = 0$ means that these joint velocities produce no instantaneous tool motion. The arm can therefore change posture while the main controller holds the tool pose.

The secondary torques are projected to preserve the main tool-pose task in the controller model. Damping opposes the extra joint velocity and becomes zero at rest. Conditioning instead depends on the joint configuration and can act even at rest. It seeks a posture with a larger minimum singular value of J , called $\sigma_{\min}$. A small $\sigma_{\min}$ means that tool motion in one direction requires large joint velocities, so increasing it helps move away from such a posture. The two torques thus address motion and posture, respectively.

Damping can therefore bring motion to rest in an unfavourable posture. Conditioning addresses the posture itself, which is why combining the two can be useful.

Slide 19 | Control OFF, then damping ON

The first video shows the effect of damping.

It starts with null-space control off, then switches the damping torque on.

[Play the full video: about 23 seconds. Then say:]

With null-space control off, the arm moves more easily. With damping on, the same push moves it more slowly, so I need to push harder to reach the same speed. Throughout this, the main controller still holds the tool pose.

Slide 20 | Conditioning ON

Conditioning has a different purpose, as this next video shows.

Here, the conditioning controller is active.

[Play the full video: about 8 seconds. Then say:]

It changes the posture while keeping the tool pose. Unlike damping, which mainly opposes motion, conditioning seeks a posture with better Jacobian conditioning.

Slide 21 | Null-space experiment

To compare these effects systematically, I applied the same disturbance in each experiment.

Joint torques represent a virtual twenty-newton force acting on joint three. I tested no null-space control, damping only, conditioning only, and both together, with three trials per case. Joint one receives the largest peak disturbance torque, so the next plots show its mean response over four seconds.

Slide 22 | Joint motion over time

First, we compare how much joint one moves.

Damping reduces the motion, while conditioning and the combined controller keep it much smaller. The zoom on the right reveals repeated direction changes. Adding damping reduces the total motion, although some reversals remain.

Slide 23 | Jacobian conditioning

We also need to check whether the resulting posture remains well conditioned.

Without null-space control, $\sigma_{\min}$ decreases, and damping alone gives the same trend. With conditioning, alone or combined with damping, it stays almost constant. This supports the distinction made earlier: damping limits motion, while conditioning helps maintain a usable posture.

Slide 24 | Conclusion

These results bring me to the main conclusions.

Allowing the tool to rotate during contact helps reduce the angular error. The CoC shift can support or oppose this alignment, while higher rotational stiffness resists it and leaves a larger error. For the arm, damping reduces unwanted motion and conditioning improves the posture response. The two can be combined.

As future work, I would improve the tool mount and adjust the CoC during alignment, returning it to the TCP afterwards. Thank you for your attention. I am happy to take your questions.

[End of the main talk. Use backups only if needed for questions.]

Use only the relevant backup when answering a question. These six scripts are outside the main talk. Each can be read on its own.

Backup B1 | Opposing moment

This extra video illustrates why the direction of the CoC shift matters.

Here, the added moment acts against alignment.

[Play the full video: about 10 seconds. Then say:]

It is the opposing case from the earlier diagrams: the shift hinders the turning motion. The result plots provide the numerical comparison.

Backup B2 | Null-space torques and projector

This slide gives the detailed equations for the two null-space torques.

The torque projector selects secondary directions that preserve the main tool-pose task in the controller model. The damping term opposes projected joint velocity, so it vanishes when the joints stop. The conditioning term instead selects the direction with a better sigma min and can act at rest. Its small deadband prevents unnecessary switching between directions.

Backup B3 | Angular quantities

This diagram distinguishes the two angular quantities used in the results.

The red arc is the angular offset at contact entry, while the blue arc is the remaining angular error. Both are measured relative to the calibrated surface. The plots show the component about the first surface tangent, so zero in that component alone does not guarantee perfect physical alignment.

Backup B4 | Baseline angular error

This baseline keeps the CoC at the TCP.

Contact reduces the two large starting offsets, but the three final errors remain around one and a half degrees. Friction, calibration and the tool mount may contribute to the remaining error. This comparison does not separate their effects.

Backup B5 | Sources of angular offset

The mismatch at contact entry can come from two sources.

The real surface may differ from the surface assumed by the controller, and the actual tool orientation may differ from its desired orientation. The green tool face and dashed desired orientation illustrate the latter. Both affect the entry angle, which is why the contact tests use a deliberate angular offset.

Backup B6 | Cumulative joint motion

Finally, this comparison adds up motion across all seven joints.

Because motion is accumulated over time, changes of direction do not cancel. Damping alone gives about twenty-five percent less motion than no null-space torque. At a conditioning setting of two newton metres, adding damping gives about fifty percent less motion than conditioning alone.